

Innovation-Normalized Conformal Prediction for Autoregressive Episodes

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ABSTRACT In statistical signal processing, prediction intervals quantify uncertainty in forecasts of future signal samples. When recordings differ in amplitude, normalization can preserve coverage, but its effect on interval length depends on temporal correlation and parameter estimation. We formulate innovation-normalized autoregressive conformal prediction (IN-ARCP) by combining an autoregressive predictor, a history-only innovation scale and split-conformal calibration. We ask how estimating the predictor and calibration threshold changes interval length when normalization removes amplitude from the score. For independent stationary Gaussian AR(1) episodes with a common coefficient, we derive the score distribution and an exact expected-length integral accounting for coefficient estimation, clipping and finite calibration. A local expansion separates fitting, calibration and quantile-rounding costs; amplitude cancels from the score but enters fitting cost through a scale-moment ratio. Numerical integration and Monte Carlo simulation agree at the evaluated configurations. Across 4,800 fitted simulation repetitions, the comparisons identify advantages and limitations: under common Gaussian dynamics, IN-ARCP gives intervals approximately 13–14% shorter than a scale-invariant Ad-EffOort adaptation at empirical coverage near 90%, while an equal-fit Student reference nearly matches it. Unmodified Ad-EffOort is shorter with unchanged amplitudes and four observations, and a nonlinear forest is shorter under varying dynamics. The main contribution is an explicit mathematical characterization of IN-ARCP, supported by numerical validation and comparative simulations.

INDEX TERMS Autoregressive processes, conformal prediction, interval estimation, prediction efficiency, scale invariance, statistical signal processing.

I. INTRODUCTION

Statistical signal prediction estimates an unobserved sample from a measured history. A prediction interval supplements the point forecast by quantifying uncertainty. This paper considers short scalar recordings collected as independent episodes: observations within each episode are temporally correlated, while the amplitude can differ between episodes. The target is the next observed sample. This setting separates uncertainty due to short histories from variation in recording amplitude and provides a tractable model for evaluating interval construction.

An autoregressive (AR) predictor exploits temporal correlation, but its residual scale must also reflect that correlation. Raw history energy combines predictable signal variation with innovation variation. Under stationary Gaussian AR(1) dynamics, an innovation transform separates these com-

ponents and yields a Student prediction pivot when the coefficient is known. In practice, the coefficient is estimated and the interval threshold is calibrated from a finite sample. These two estimation steps determine the length of the resulting interval.

A. RESEARCH PROBLEM AND QUESTION

Normalized split conformal prediction provides a general coverage construction [1]. Existing expected-size identities [2] and efficiency analyses [3], [4] also establish that training and calibration affect interval length. These general results do not directly evaluate the distribution of the clipped pooled AR estimator, its induced score distribution and the resulting expected length for the construction considered here. The problem addressed in this paper is to calculate that model-specific relationship, including the effect of

amplitude heterogeneity in training.

The research question is: *how does history-innovation normalization affect prediction coverage and expected interval length when the AR coefficient and calibration threshold must both be estimated?* We examine the hypothesis that a homogeneous innovation scale removes episode amplitude from the prediction score while leaving an amplitude-dependent cost in coefficient estimation. A second, empirical question is when this model-based scale gives shorter intervals than raw-energy, learned-scale or quantile-based alternatives at comparable coverage.

B. CONSTRUCTION AND CONTRIBUTIONS

We formulate innovation-normalized autoregressive conformal prediction (IN-ARCP). A clipped pooled AR coefficient defines the center; a history-only innovation norm defines the residual scale; and a separate calibration set defines the conformal quantile. The construction specializes normalized split conformal prediction to the AR episode model. Its mathematical characterization and simulation evaluation comprise the following contributions:

- 1) A derivation of the innovation normalization and a scale-conditioned coverage result under exchangeable standardized episodes. Under the stationary Gaussian model, we derive the fitted-coefficient score distribution and an exact integral for expected interval length, including coefficient clipping and finite calibration.
- 2) An explicit asymptotic expansion of excess expected length into coefficient-fitting, calibration-variability and quantile-rounding terms. The fitting term identifies the role of the fourth-to-second moment ratio of training amplitudes, even though amplitude cancels from the score.
- 3) Numerical evaluation of the analytical expressions and paired simulations against raw-energy ablations, learned conformal intervals, a mixture-quantile implementation and a Gaussian Student reference. These experiments assess the hypothesis under common dynamics and identify its limitations under changing amplitudes or dynamics.

The coverage argument uses established conformal rank theory; the analytical contribution concerns the specific score and estimator laws and their effect on expected length. Section III derives the construction, Sections IV and V develop the analysis, and Sections VI and VII report the simulation design and results.

II. RELATED WORK

The closest literature concerns normalized prediction scores, expected interval length and learned interval construction. These works address different parts of the research question: coverage after normalization, the cost of finite calibration, and the effect of fitting a predictor. The comparison below identifies the quantity analyzed in each case before relating it to IN-ARCP.

A. NORMALIZATION, COVERAGE AND EFFICIENCY

Lei et al. [1] develop distribution-free regression intervals, including locally normalized split-conformal scores. Setting their regression estimate to an autoregressive center and their residual scale to the innovation norm gives the interval studied here. Dewolf et al. [5] examine conditional validity in heteroskedastic conformal regression, providing related context for pivotal normalization. The coverage result below specifies exactly what is conditioned on: the fitted model and episode scales, while calibration randomness and the standardized test episode are averaged out. It does not provide coverage conditional on every realized history.

Dhillon et al. [2] derive general expected-size expressions for conformal sets. Our beta-integral expression evaluates this type of order-statistic relationship for an autoregressive score and couples it to the training-estimator law. Yao, He and Gastpar [4] analyze nonasymptotic efficiency of conformalized regression under a different training setup, including bounded observations and a compact linear class. The Gaussian model here is unbounded, and our explicit local expansion is not a stronger distribution-free inequality. Asymptotic comparison of prediction-interval lengths also has substantial earlier precedent, including autoregressive examples in Kabaila and Syuhada [6].

The analytical problem here is the expected length of an innovation-normalized interval with a clipped pooled AR coefficient. The expected-size identity in [2, Th. 1] applies broadly once the score distribution is specified. Here that distribution is evaluated for a fitted autoregressive coefficient and then integrated over its finite-training law. The bounds in [4, Ths. 3.2 and 4.1] concern expected absolute deviation from conditional oracle length for quantile or median regressors. Our expansion concerns mean length relative to a scale-equivariant oracle at fixed coverage level. Its explicit coefficients expose the effect of training-scale moments and integer calibration ranks; they are not a rate improvement over a theorem for the same target and model.

B. LEARNED INTERVALS AND INVARIANCE

Le Bars and Humbert [3] analyze excess interval length for split conformal regression and EffOrr, relating predictor fitting to calibration. Their Ad-EffOrr extension learns intervals whose widths depend on the covariates. Thus training and calibration costs are already studied in this literature. The present analysis evaluates these costs for the specified autoregressive estimator through its score distribution, finite-training law and local expansion. The experiments include a linear Ad-EffOrr implementation and a separately identified scale-invariant adaptation.

Conformalized quantile regression (CQR) [7] calibrates estimated lower and upper conditional quantiles. Our mixture-quantile comparator uses this calibration rule with a fitted Gaussian scale-mixture model; it does not reproduce the quantile learners evaluated in [7]. CD-split and HPD-split [8] instead construct regions from estimated conditional

densities. Neither density-based method is implemented in the present comparisons.

Symmetry-based inference has broader formulations, including SymmPI [9], conditional structured-data inference [10], and standardized multivariate residuals [11]. Those frameworks address structured-data or multivariate coverage. Here, positive episode scales cancel algebraically from a scalar residual score. Invariant probabilistic prediction [12] studies a different predictive objective. The present coverage result conditions on episode amplitudes and averages over the standardized histories; it does not establish the structured-data or multivariate conditional guarantees studied in those papers.

Classical best-invariant interval theory [13] motivates the oracle comparison developed below. The derivation uses standard radial independence and a Student pivot. The numerical evaluation uses Gaussian quadratic-form inversion [14].

C. PREDICTION TARGETS AND ANALYTICAL SCOPE

Recent conformalized Kalman-filter work [15] uses filter moments to construct uncertainty regions for latent states, with sample-wise and trajectory-wise calibration across exchangeable trajectories. Our target is the next observed scalar from a short history, not a latent state or an entire trajectory. A numerical comparison would require a common observation model and target. Likewise, selecting among fitted predictors is a separate problem: data-dependent selection can alter conformal validity and requires an appropriate procedure [16].

Table 1 compares the analytical results most directly related to this paper. The implemented predictors and their calibration rules are specified in Section VI; their numerical results are reported in Section VII.

III. IN-ARCP: MODEL AND DERIVATION

A. INDEPENDENT EPISODES AND INFORMATION SETS

An episode is $X = (H_1, \dots, H_m, Y) \in \mathbb{R}^{m+1}$, where H is observed before predicting Y , and $m \geq 2$. Use disjoint sets of N training episodes and n calibration episodes, together with independent evaluation episodes. Within-episode observations can be dependent. Independence is imposed across episodes, not on all scalar measurements after pooling them.

For the coverage argument, write $X_i = \sigma_i Z_i$, with $\sigma_i > 0$. Conditional on the training data and all calibration/test scales, the standardized calibration and test vectors must be exchangeable. Independent common-law standardized episodes independent of their scales suffice. Their distribution need not be Gaussian for this argument.

Exact performance calculations use the stronger model

$$\begin{aligned} X_1 &| \sigma \sim \mathcal{N}(0, \sigma^2), \\ X_{j+1} &= \rho X_j + \varepsilon_{j+1}, \quad j = 1, \dots, m, \\ \varepsilon_{j+1} &| \sigma \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2(1 - \rho^2)). \end{aligned} \quad (1)$$

The initial state and innovations are conditionally independent, and $\rho \in (-1, 1)$ is common across episodes. The marginal scale σ differs from the innovation scale $\tau = \sigma\sqrt{1 - \rho^2}$. Zero mean is an assumption of the model; it is not obtained by centering an episode using its unobserved response.

B. COEFFICIENT ESTIMATION

Let $X_{i,m+1} = Y_i$ on the training set and choose $a = 0.98$. Fit

$$\tilde{\rho} = \frac{\sum_{i=1}^N \sum_{j=1}^m X_{ij} X_{i,j+1}}{\sum_{i=1}^N \sum_{j=1}^m X_{ij}^2}, \quad \hat{\rho} = \text{clip}(\tilde{\rho}, -a, a). \quad (2)$$

The sum includes transitions from the last training-history observation to its observed response. The coefficient is fixed before calibration.

C. DERIVATION OF THE INNOVATION SCALE

Under (1), define $U_1 = \sqrt{1 - \rho^2} H_1$ and $U_j = H_j - \rho H_{j-1}$ for $j = 2, \dots, m$. These are independent Gaussian variables with common variance $\tau^2 = \sigma^2(1 - \rho^2)$. Their sum of squares estimates the innovation variance:

$$\frac{\sum_{j=1}^m U_j^2}{\tau^2} \sim \chi_m^2, \quad \mathbb{E} \left[\frac{1}{m} \sum_{j=1}^m U_j^2 \mid \sigma \right] = \tau^2. \quad (3)$$

Replacing ρ by a fitted coefficient r gives

$$s_r(h)^2 = \frac{(1 - r^2)h_1^2 + \sum_{j=2}^m (h_j - rh_{j-1})^2}{m}. \quad (4)$$

The factor $1 - r^2$ makes the initial observation commensurate with the subsequent innovations at $r = \rho$. For an arbitrary fitted r , the resulting scale is positive and homogeneous in the history, but need not be unbiased for τ or τ^2 . The training denominator and history scale are assumed positive almost surely.

D. CONFORMAL CALIBRATION AND PREDICTION

The future residual $Y - \rho H_m$ is independent of the Gaussian history innovations. Consequently $(Y - \rho H_m)/s_\rho(H)$ has a Student t_m distribution. IN-ARCP replaces this known-coefficient threshold by an empirical quantile of residuals computed with the fitted coefficient.

With $p = 1 - \alpha$, define calibration scores and the rank index

$$R_i = \frac{|Y_i - \hat{\rho} H_{im}|}{s_{\hat{\rho}}(H_i)}, \quad k = \lceil (n+1)p \rceil, \quad \hat{q} = R_{(k)}. \quad (5)$$

If $k > n$, set $\hat{q} = \infty$. The interval is

$$I(h) = [\hat{\rho} h_m - \hat{q} s_{\hat{\rho}}(h), \hat{\rho} h_m + \hat{q} s_{\hat{\rho}}(h)]. \quad (6)$$

The interval is the inversion of $|y - \hat{\rho} h_m|/s_{\hat{\rho}}(h) \leq \hat{q}$; its center and scale depend only on the observed history.

Implementation proceeds by fitting (2) once, computing (5) on the calibration episodes, retaining the specified order

TABLE 1. Comparison with prior analyses of conformal coverage and interval length.

Reference	Established result	Result evaluated here
Lei et al. [1]; Dewolf et al. [5]	Normalized residual intervals and conditions for marginal or conditional validity	Coverage conditional on episode scales when standardized episodes are exchangeable (Proposition 1)
Dhillon et al. [2]	General expected-size expressions for split-conformal prediction sets	Mean length obtained by integrating the AR score quantile over calibration and coefficient-estimation distributions ((13), (16))
Le Bars and Humbert [3]	Excess-length bounds and predictor fitting directed at interval length; adaptive Ad-EffOrt intervals	Exact mean and local expansion for a specified pooled AR estimator, rather than optimization over a learned predictor class
Yao et al. [4]	Nonasymptotic bounds on deviation from oracle length, accounting for training size, calibration size and miscoverage level	Fixed-coverage asymptotic mean expansion with explicit scale-moment and quantile-rounding terms (Theorem 1)

statistic, and evaluating (6) for each history. Computing the pooled coefficient costs $O(Nm)$ arithmetic, calibration-score evaluation costs $O(nm)$, and a new interval costs $O(m)$. Selecting a quantile can be performed without a full sort. Execution times were not measured.

Table 2 distinguishes the premises of the coverage guarantee from the stronger assumptions used to calculate efficiency.

IV. COVERAGE AND EXACT PERFORMANCE ANALYSIS

A. SCALE-TRANSFER COVERAGE

Proposition 1. *Under the standardized-score exchangeability premise in Section III,*

$$\mathbb{P}\{Y_* \in I(H_*) \mid T, \sigma_1, \dots, \sigma_n, \sigma_*\} \geq p, \quad (7)$$

where T is the training data. If scores have no ties and $k \leq n$, the conditional coverage is $k/(n+1)$.

Proof. For every $c > 0$, $s_r(ch) = cs_r(h)$ and $r(ch_m) = c(rh_m)$. Thus the scale cancels in (5). Conditional on T , the coefficient is fixed and all calibration/test scores use the same function of exchangeable standardized episodes. Their ranks are exchangeable. Comparing the test score with the corrected calibration order statistic proves the claim, with the usual conservative treatment of ties. $\square$

This is a standard conformal rank argument specialized to the score. It averages over calibration and the standardized test episode. It is not a guarantee for each realized calibration set or each realized history. Whole-episode scaling preserves the score; a change in only the future response, or a change in standardized dynamics, need not do so.

B. SCORE DISTRIBUTION FOR AN ARBITRARY FITTED COEFFICIENT

Under (1), let W_r be lower bidiagonal with first diagonal entry $\sqrt{1-r^2}$, remaining diagonal entries one, and subdiagonal entries $-r$. Write $L = W_\rho^{-1}$ and $Q_r = W_r^T W_r$. Then $U = W_\rho H$ and $E = Y - \rho H_m$ comprise $m+1$ independent $\mathcal{N}(0, \tau^2)$ coordinates.

Set $U = RD$, where D is uniform on the unit sphere in $\mathbb{R}^m$, $R/\tau \sim \chi_m$, and R and D are independent. The variable $V = E/(R/\sqrt{m})$ has the Student t_m distribution and is independent of D . For a coefficient r fitted independently of this episode, define

$$b_r(D) = \sqrt{D^T L^T W_r^T W_r L D}, \quad (8)$$

$$a_r(D) = \sqrt{m}(r - \rho)e_m^T L D. \quad (9)$$

The score event is $|V - a_r(D)| \leq qb_r(D)$. Therefore its cumulative distribution function (CDF) is

$$G_r(q) = \mathbb{E}_D\{F_m(a_r(D) + qb_r(D)) - F_m(a_r(D) - qb_r(D))\}, \quad (10)$$

where F_m is the Student CDF and $q \geq 0$. This law is independent of σ . At $r = \rho$, it becomes $G_\rho(q) = 2F_m(q) - 1$. Away from ρ , both normalized center displacement and the direction-dependent scale affect the distribution.

The quadratic-form representation used for numerical evaluation is given in Appendix B.

C. FINITE-CALIBRATION EXPECTED LENGTH

For the finite-mean formulas in this and the following subsection, assume $k \leq n$; otherwise the interval and its mean length are infinite. Assume independent common-law standardized calibration episodes, not merely exchangeability. Conditional on r , continuity gives

$$\hat{q} \stackrel{d}{=} G_r^{-1}(B), \quad B \sim \text{Beta}(k, n+1-k). \quad (11)$$

Let $J(r) = \mathbb{E}_D b_r(D)$ and

$$A_m = \sqrt{2/m} \frac{\Gamma((m+1)/2)}{\Gamma(m/2)}. \quad (12)$$

For fixed positive test scale σ , the conditional expected length is

$$\begin{aligned} \ell_n(r; \sigma) &= 2\sigma \sqrt{1-\rho^2} A_m J(r) \\ &\times \int_0^1 G_r^{-1}(u) \beta_{k, n+1-k}(u) du. \end{aligned} \quad (13)$$

TABLE 2. Assumptions for coverage, exact mean length and the asymptotic expansion.

Conclusion	Required structure	Limitations
Scale-conditioned coverage	Exchangeable standardized calibration/test episodes conditional on training and scales; positive history scale	Averages calibration and history; no per-history guarantee
Exact score and mean laws	Independent stationary zero-mean Gaussian episodes; common coefficient; finite calibration rank	Changes in means, dynamics or observation noise need new laws
Known-parameter oracle	Same Gaussian model; homogeneous endpoints; independent auxiliary information	Optimal only within the stated scale-equivariant interval class
Local efficiency expansion	Fixed history length and coverage; interior coefficient; iid training scales independent of shape with a finite $4 + \eta$ moment	Asymptotic mean expansion; no finite remainder bound

Here $\beta_{a,b}$ denotes the beta density. Independence of calibration and test histories permits factorization of the threshold and test-scale expectations. Radial independence gives the remaining $A_m J(r)$ factor. This is an exact integral characterization, not an elementary closed form. For $m \geq 2$ and $k \leq n$, the mean is finite: the score has polynomial tail order m , and $n + 1 - k \geq 1 > 1/m$ controls the order-statistic mean.

D. FINITE-TRAINING UNCERTAINTY AND THE COMPLETE MEAN

The preceding mean conditions on the fit. To average over training, first condition on its positive scales $\sigma = (\sigma_1, \dots, \sigma_N)$. Let C be the Cholesky factor of the full-episode correlation matrix, whose entries are $\rho^{|j-l|}$. Let A have $1/2$ on its first upper and lower diagonals and zero elsewhere, and let $B_0 = \text{diag}(1, \dots, 1, 0)$. If $\lambda_l(x)$ are the eigenvalues of $C^T(A - xB_0)C$, positivity of the training denominator gives

$$\{\tilde{\rho} \leq x\} = \left\{ \sum_{i=1}^N \sum_{l=1}^{m+1} \sigma_i^2 \lambda_l(x) Z_{il}^2 \leq 0 \right\}, \quad (14)$$

with independent standard normal Z_{il} . Thus

$$\begin{aligned} \psi_x(t \mid \sigma) &= \prod_{i=1}^N \prod_{l=1}^{m+1} (1 - 2it\sigma_i^2 \lambda_l(x))^{-1/2}, \\ F_N(x \mid \sigma) &= \frac{1}{2} - \frac{1}{\pi} \int_0^\infty \frac{\text{Im} \psi_x(t \mid \sigma)}{t} dt. \end{aligned} \quad (15)$$

The clipped estimator has atoms $F_N(-a)$ at $-a$ and $1 - F_N(a)$ at a . Consequently,

$$\begin{aligned} \mathbb{E}[I(H_*) \mid \sigma, \sigma_*] &= \ell_n(-a; \sigma_*) F_N(-a \mid \sigma) \\ &\quad + \int_{(-a,a)} \ell_n(r; \sigma_*) dF_N(r \mid \sigma) \\ &\quad + \ell_n(a; \sigma_*) [1 - F_N(a \mid \sigma)]. \end{aligned} \quad (16)$$

Integration with respect to the coefficient-estimator distribution retains the endpoint masses caused by clipping, as well as the continuous interior contribution. With random independent training scales, average the conditional law over their specified distribution. With independent random test scale of finite mean, averaging is immediate because ℓ_n is

linear in σ_* . The numerical study checks the component distributions and one complete fitted-mean case conditional on fixed training scales. It does not integrate over a random training-scale distribution.

V. ORACLE BENCHMARK AND EFFICIENCY DECOMPOSITION

A. KNOWN-PARAMETER SCALE-EQUIVARIANT BENCHMARK

Let $c = F_m^{-1}(1 - \alpha/2)$. For known ρ and population calibration, the Student interval has expected length

$$L_* = 2c\sigma_* \sqrt{1 - \rho^2} A_m. \quad (17)$$

Proposition 2. *Under (1), consider measurable intervals whose endpoints are homogeneous of degree one in the test history. They may depend on independent auxiliary information whose law is fixed as the test scale varies. Among intervals with marginal coverage at least p , their expected length is at least L_* . The interval $\rho h_m \pm c_\rho(h)$ attains equality.*

Proof. After subtracting ρH_m and dividing by $R/\sqrt{m}$, homogeneity makes the residual endpoints functions only of D and auxiliary information. Let their nonnegative separation be L , with empty intervals assigned length zero. The Student pivot is independent of those variables. For fixed L , symmetry and strict unimodality maximize its interval probability at centered endpoints. Hence coverage is at most $\mathbb{E}g(L)$, where $g(l) = 2F_m(l/2) - 1$. This increasing function is strictly concave, so $p \leq \mathbb{E}g(L) \leq g(\mathbb{E}L)$ implies $\mathbb{E}L \geq 2c$. Independence of R gives $\mathbb{E}|I| = \mathbb{E}(R/\sqrt{m})\mathbb{E}L \geq L_*$. Equality requires the centered constant-width normalized interval almost surely. $\square$

The bound is an oracle result within the stated class, consistent with classical invariant prediction principles [13]. It is not finite-training optimality or a lower bound for every non-equivariant population-calibrated procedure. Comparators using a different validity target can have shorter average intervals without contradicting it.

B. EXPLICIT LOCAL EXPANSION

Define $d = 1 - \rho^2$, $f = f_m(c)$, and the scale heterogeneity factor

$$\kappa = \frac{\mathbb{E}\sigma^4}{(\mathbb{E}\sigma^2)^2}. \quad (18)$$

The scalar coefficients needed below are

$$V_u = \frac{(m-1) - (m-3+2/m)\rho^2}{m(m+2)d^2}, \quad (19)$$

$$K = \frac{m+1}{2(m+c^2)} \left(c^2 V_u + \frac{1}{d} \right), \quad (20)$$

$$V_\rho = \frac{d\kappa}{m}, \quad C_m = \frac{p(1-p)(m+1)}{8(m+c^2)f^2}. \quad (21)$$

Theorem 1. *Assume independent Gaussian standardized episodes, iid positive training scales independent of those shapes, and $\mathbb{E}\sigma^{4+\eta} < \infty$ for some $\eta > 0$. Fix $m \geq 2$, $\alpha \in (0, 1)$, and ρ strictly inside the clipping interval. With $\epsilon_n = k - (n+1)p \in [0, 1]$, as $N, n \rightarrow \infty$, at fixed positive test scale,*

$$\begin{aligned} \frac{\mathbb{E}|I|}{L_*} &= 1 + \frac{KV_\rho}{N} + \frac{C_m}{n+2} \\ &\quad + \frac{\epsilon_n}{2cf(n+1)} + o(N^{-1} + n^{-1}). \end{aligned} \quad (22)$$

The proof is given in Appendix A. The first term in excess length arises from learning the coefficient, the second from finite calibration, and the third from rounding the rank index. The expression is a local asymptotic expansion; it supplies no explicit finite-sample remainder bound. It is not uniform near a unit root, for growing history length, or as α tends to zero.

Although σ disappears from the conformal score, κ remains in the variance of the pooled estimator. Therefore scale-transfer coverage and scale-insensitive learning efficiency are distinct properties. For $\rho = 0.8$, $\alpha = 0.1$, $\kappa = 1.5$, and $N = 1000$, the leading fitting costs are approximately 0.0227% for $m = 4$ and 0.0068% for $m = 12$. At $n = 1999$, the corresponding leading calibration costs are approximately 0.104% and 0.067%. The calibration terms predict a small length difference between conformal calibration and the Student threshold when both use the same fitted coefficient. The fitting term instead measures part of the difference from the known-coefficient oracle.

VI. EXPERIMENTAL DESIGN

A. OBJECTIVES, GENERATORS AND REPLICATION

The experiment separates three objectives: checking the analytical components, measuring efficiency at comparable coverage, and testing changes that violate the coverage premise. Following explicit simulation-design principles [17], independent fitted repetitions are the unit used for Monte Carlo uncertainty.

The main comparison uses $N = 1000$ training episodes, $n = 1999$ calibration episodes, and 2000 evaluation episodes per fitted repetition, with $m \in \{4, 12\}$ and $\alpha = 0.1$. Fifty development repetitions per cell precede

200 confirmation repetitions. Nine scenarios give 18 confirmation cells. Unless otherwise specified, episode variance is $\sigma^2 = 3/G$, $G \sim \text{Gamma}(4, 1)$, independently of standardized shapes; this gives $\mathbb{E}\sigma^2 = 1$ and $\kappa = 1.5$.

The nine scenarios are: independent Gaussian observations with unit scale; independent Gaussian observations with the preceding random scale; an equal-probability variance mixture on $\{0.1, 4\}$; independent standardized Student t_5 observations; common Gaussian AR(1) with $\rho = 0.8$; independent Gaussian episodes with doubled test scale; AR(1) with doubled test scale; an AR coefficient change from 0.8 to -0.4 at test time; and a future-only multiplication $Y \mapsto 2Y$. Student coordinates are standardized to unit variance. The last two changes violate common standardized shape. Whole-episode doubling multiplies both history and response.

A separate challenge uses three mechanisms and both history lengths, again with 200 fitted repetitions per cell. They are AR(1) with $\rho = 0.8$ and standardized t_5 innovations; Gaussian AR(2) with coefficients $(0.5, 0.3)$; and Gaussian AR(1) with an independently selected episode coefficient $\rho_i \in \{-0.8, 0.8\}$ of equal probabilities. The first two use 256 discarded initialization steps. The AR(2) innovation variance is 0.4457142857, giving unit stationary marginal variance. All challenge test episodes have twice the training/calibration amplitude. This is genuine innovation non-Gaussianity in the first challenge, rather than merely an episode-wide Gaussian scale mixture.

There are 4800 independent fitted confirmation/challenge repetitions in total. Invariant-adaptation and classical-reference comparisons reuse paired episodes and do not increase this count. Separate random streams generate training, calibration, evaluation and model randomness. Random-number configurations are provided in the reproducibility supplement.

B. COMPARATORS AND THEIR INFORMATION

All fitted comparisons use the same episode splits and observed history length. The basic ablations are zero-centered raw-root-mean-square (RMS) normalization, an AR center with raw RMS, and an AR center with a globally calibrated unnormalized residual. They distinguish the contribution of predicting the center from that of estimating the history scale.

The locally weighted comparator uses an ordinary least-squares (OLS) center and a random-forest (RF) residual scale, followed by normalized absolute-residual calibration. Its feature vector contains history coordinates and history RMS. The scale forest uses 100 trees and depth five. The linear Ad-Effort implementation uses 1000 optimization steps, the published smoothing/step-size form, and a 100-tree depth-five quantile forest at quantile 0.9. The optimizer follows the authors' implementation [18]; the source revision is recorded in the supplement. Its conformal adjustment is additive to the fitted radius, whereas the locally weighted comparator uses multiplicative normalization. The repro-

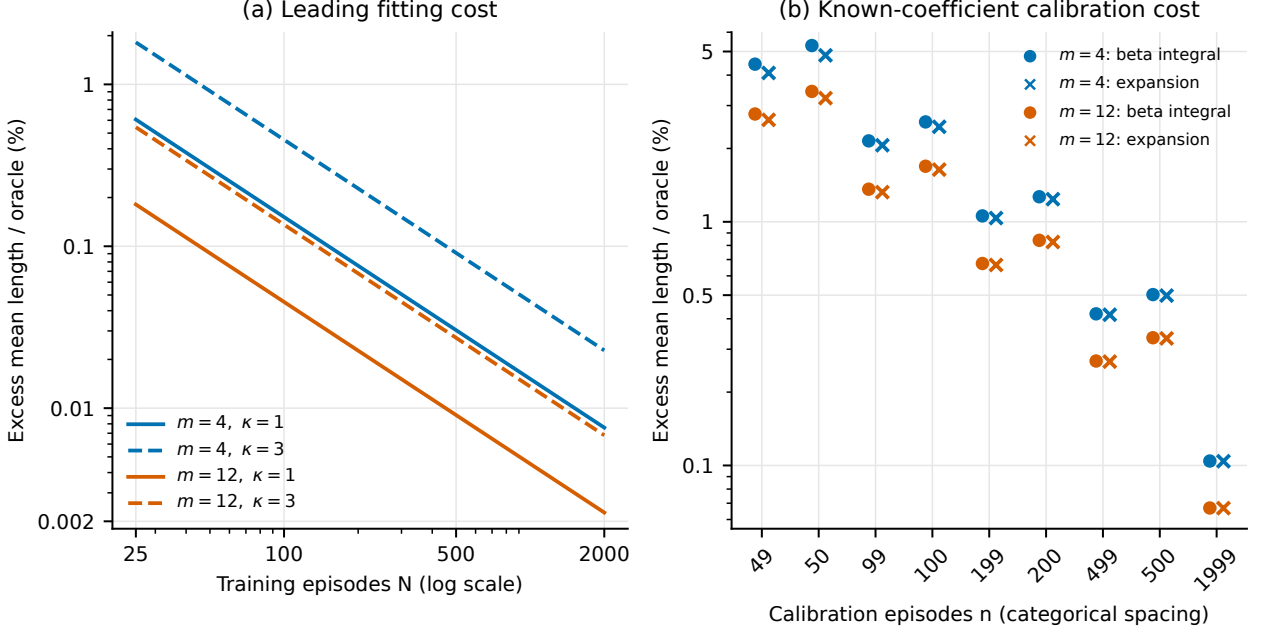


FIGURE 1. Analytical efficiency components at $\rho = 0.8$ and $\alpha = 0.1$, expressed as excess mean length relative to the known-parameter oracle. (a) Leading fitting term in (22); both axes are logarithmic. These are analytical predictions, not simulated finite-sample results. (b) Known-coefficient calibration overhead from numerical beta integration and the expansion including rounding. The vertical axis is logarithmic; calibration sizes have categorical spacing to separate adjacent ranks. Blue denotes history length $m = 4$, orange $m = 12$; $\kappa = \mathbb{E}\sigma^4 / (\mathbb{E}\sigma^2)^2$ measures scale heterogeneity.

ducibility supplement specifies software versions, optimizer settings and source provenance.

The mixture-quantile comparator fits a zero-centered Gaussian scale-mixture model and calibrates its central quantile interval using the conformalized-quantile principle [7]. It is included as a specified model-based instantiation, not as a complete reproduction of a published CQR benchmark.

For the matched-invariance comparison, let $A(H) = \sqrt{m^{-1} \sum_j H_j^2}$. Fit and calibrate the comparator on $(H/A(H), Y/A(H))$, then multiply its test endpoints by $A(H_*)$. This transformation is our adaptation. The mathematically constant normalized RMS feature is omitted for OLS or represented exactly as a constant for Ad-EffOrrt. This avoids numerical variation in a redundant feature. Native and adapted methods remain separately identified.

The nonlinear challenge comparator uses an RF center with 200 trees, depth eight and minimum leaf size ten, together with a 100-tree depth-five residual-scale forest with minimum leaf size ten. It receives the same normalized history information. The Gaussian plug-in reference uses the same fitted coefficient and history scale as (6), replacing only $\hat{q}$ by $F_m^{-1}(0.95)$. It is a useful equal-fit reference, but it does not inherit the conformal coverage guarantee under arbitrary standardized distributions.

C. PERFORMANCE MEASURES AND UNCERTAINTY

For each fitted repetition b , estimate coverage and mean interval length over its evaluation episodes. Average these

quantities over repetitions. For paired mean lengths L_b and M_b , report the ratio $\hat{R} = L/M$, with delta-method Monte Carlo standard error

$$\widehat{\text{se}}(\hat{R}) = \frac{\text{sd}(L_b - \hat{R}M_b)}{\sqrt{B} \bar{M}}. \quad (23)$$

Coverage standard errors likewise use the between-repetition standard deviation divided by $\sqrt{B}$. Evaluation episodes sharing a fitted/calibrated model are not treated as independent fitted models.

The numerical comparison criterion requires coverage lower bounds above 0.895 for both methods and a length-ratio upper bound below 0.95. Bounds use 3.890592 Monte Carlo standard errors, the normal critical value corresponding to a two-sided 5% Bonferroni calculation for 1000 comparisons. This was the fixed screening budget in the simulation protocol, not the number of entries in the tables. The normal approximation and the absence of a complete accounting of all exploratory comparisons preclude a confirmed familywise error guarantee. The coverage criterion permits a shortfall of 0.005 from the nominal 0.90 level; it is not a test of exact nominal coverage. Ratio intervals in the principal comparison plots use this multiplier; coverage error bars use 1.96 standard errors. A short interval with inadequate coverage is not interpreted as an efficiency improvement. Hyperparameters are fixed for the reported runs; comprehensive equal-budget tuning is not established.

VII. RESULTS

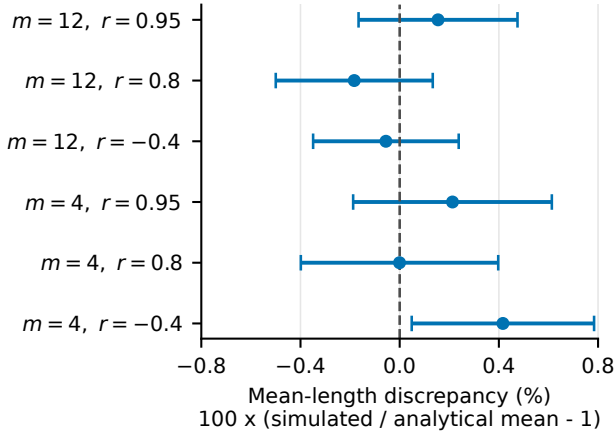


FIGURE 2. Direct finite-calibration mean-length simulation versus (13). The horizontal axis is percentage discrepancy, with symmetric limits about zero. Bars are approximate 95% Monte Carlo intervals from independent repetitions, not uncertainty bands for the mathematical identity. Each row fixes history length m and fitted coefficient r .

A. ACCURACY OF THE EXPECTED-LENGTH CALCULATION

The score CDF was evaluated by quadratic-form inversion and compared with Monte Carlo estimates from Gaussian episodes generated using a covariance Cholesky factor. Across 36 score-CDF checks against 300000 episodes per case, the maximum discrepancy is 3.03 estimated Monte Carlo standard errors. Six finite-calibration mean checks simulate both calibration and test histories, with 2000 repetitions and 512 test histories per repetition; the maximum discrepancy is 2.22 standard errors (Fig. 2). The training CDF is checked at 32 points over eight settings using 60000 independent training sets per setting, with maximum discrepancy 1.93 standard errors. Four additional checks examine the clipping probabilities.

Eight deterministic curvature comparisons give a maximum relative discrepancy of approximately 0.0341%. A further 25 matrix checks span $m \in \{2, 3, 4, 12, 25\}$ and $\rho \in \{-0.95, -0.4, 0, 0.4, 0.95\}$, verifying whitening, the precision determinant and the closed-form directional variance in (19). The evaluation grid and numerical tolerances are listed in the reproducibility supplement.

For a complete check of (16), take $m = 2$, $\rho = 0.4$, $N = 2$, fixed training scales (0.7, 1.3), $n = 19$, and unit test scale. Numerical Stieltjes integration, retaining both clipping atoms, gives mean length 6.9372. Monte Carlo simulation with 30000 fitted repetitions and 32 test episodes per repetition gives 6.9425 with Monte Carlo standard error 0.02579, a discrepancy of 0.205 standard errors. Coverage is 0.90010 with standard error 0.00048. Increasing the outer quadrature from 64 to 128 bins changes the mean by 0.000675; doubling angular and beta quadrature resolutions at the true coefficient and clipping endpoints changes their conditional means by at most 0.00121. This checks one small-history configuration with fixed training scales, separately from the 4800-repetition comparator study.

At the true coefficient, numerical beta integration supplies a direct check of the calibration component in Fig. 1. The adjacent values of n show why integer quantile rounding should not be discarded. The fitting curves in that figure are analytical leading terms. The training-size and heterogeneity curves represent the asymptotic approximation.

B. OVERVIEW OF THE IMPLEMENTED COMPARISONS

Figure 3 reports nine comparator configurations for the common Gaussian AR(1) scenarios: three ablations, unmodified OLS–RF normalized regression, unmodified linear Ad-EffOrt, the mixture-quantile construction, two scale-invariant adaptations and the equal-fit Student reference. Each comparison uses the same training, calibration and evaluation episodes as IN-ARCP. The nonlinear RF-center comparison under varying dynamics is reported separately in Fig. 5.

Without an amplitude change, global residual calibration and unmodified learned methods can produce shorter intervals than IN-ARCP at coverage near 90%. After the test amplitudes double, several of these methods lose coverage, whereas the history-normalized methods retain coverage near the nominal level. Thus the comparison distinguishes average length under an unchanged population from validity under a change in episode amplitude. The complete saved method-level results are included in the supplementary CSV files.

C. COMMON DYNAMICS AND MATCHED-INVARIANCE COMPARISON

For common Gaussian AR(1) episodes, the length ratios relative to the invariant Ad-EffOrt adaptation are 0.863 and 0.869 for $m = 4$ and 12. With doubled test amplitude they are 0.866 and 0.867. Both methods retain coverage close to 0.90; Table 3 reports the actual estimates and paired ratio intervals. All four comparisons satisfy the numerical criterion in Section VI.

With the same AR center, replacing raw RMS by innovation RMS gives ratios of 0.832 and 0.891. This ablation supports the relevance of accounting for temporal dependence in the history scale. The benefit is not equivalent to superiority over an unrestricted fitted interval. Against native linear Ad-EffOrt in the unchanged AR(1) setting, the ratios are 1.164 and 1.005: the analyzed interval is longer for $m = 4$, and the $m = 12$ comparison is close. Fig. 4 retains both native and adapted comparisons. The reduction relative to the scale-invariant adaptation therefore does not imply a reduction relative to unmodified Ad-EffOrt.

D. CLASSICAL REFERENCE AND MODEL MISSPECIFICATION

The Gaussian plug-in Student reference essentially matches the analyzed method under the Gaussian AR model. Their length ratios are approximately 0.998 and 1.002, with coverage near 0.90. Because both procedures use the same

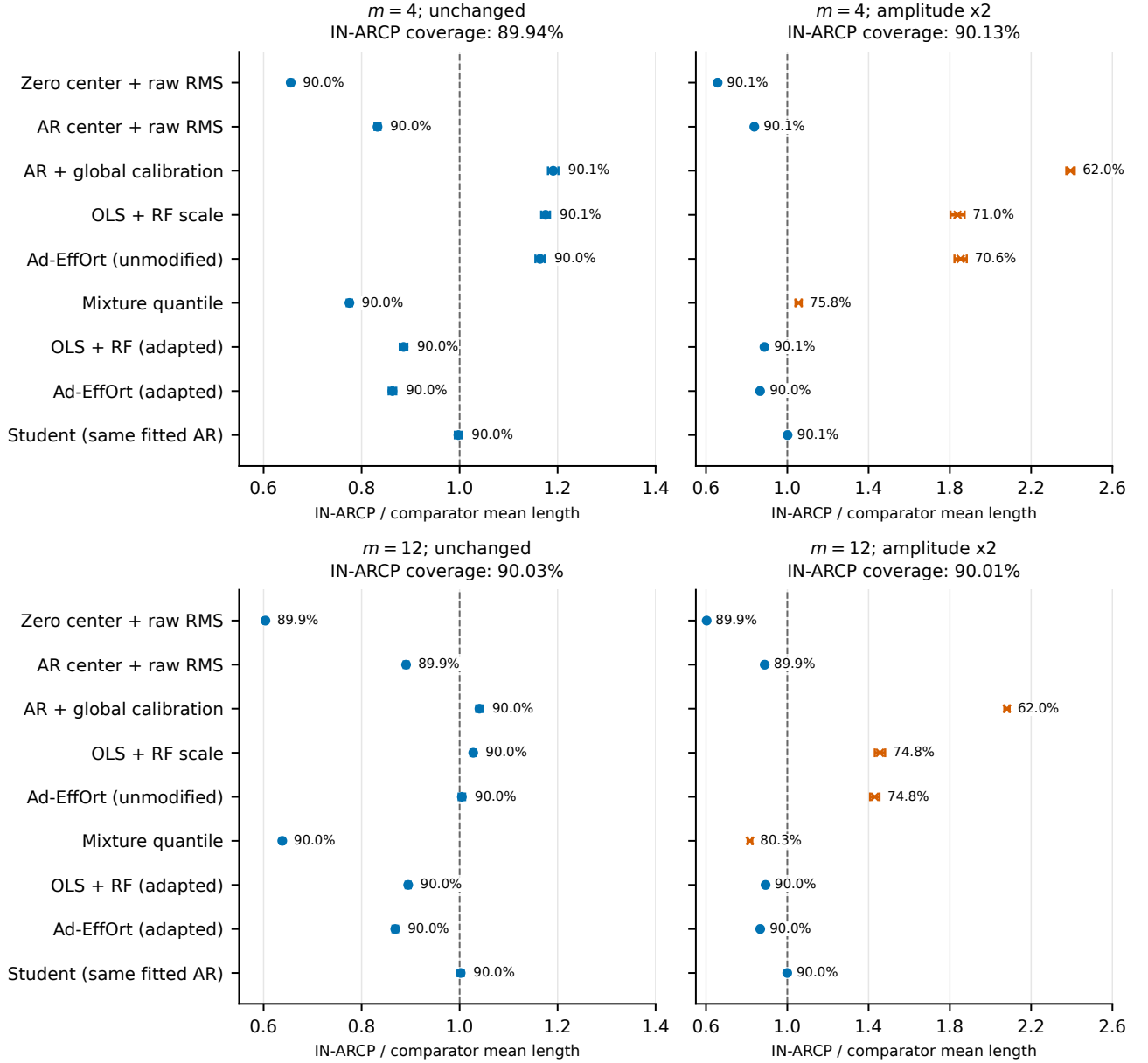


FIGURE 3. Common Gaussian AR(1) comparisons for histories of length 4 (top) and 12 (bottom), with unchanged test amplitudes (left) and doubled test amplitudes (right). Ratios are IN-ARCP mean length divided by comparator mean length; values below one favor IN-ARCP in length. Bars show paired Monte Carlo intervals using multiplier 3.890592. Numbers beside the markers give comparator coverage in percent; IN-ARCP coverage is stated in each panel. Orange crosses identify empirical comparator coverage below 89.5%, so a shorter interval in those cases is not a comparable-coverage efficiency advantage. The unchanged-amplitude panels use a narrower horizontal range. Adaptations are labeled explicitly.

fitted coefficient, this comparison isolates the calibration-threshold difference; it does not measure coefficient-estimation cost relative to the oracle. Under genuinely heavy-tailed innovations, the plug-in reference undercovers in the reported settings, whereas conformal calibration yields coverage near 0.90 at the cost of greater length (Table 4).

Gains against the invariant linear Ad-EffOrit adaptation persist in the heavy-tailed AR(1) and Gaussian AR(2) challenges: ratios are approximately 0.866/0.864 and

0.916/0.918 for $m = 4/12$. However, the varying-coefficient challenge reverses the efficiency ranking. Relative to the nonlinear RF-center comparator, the IN-ARCP ratios are 1.186 and 1.308. For $m = 12$, coverage is approximately 0.8996 versus 0.8992, while mean lengths are approximately 6.674 versus 5.103. The RF-center procedure retains its advantage at comparable coverage (Fig. 5).

TABLE 3. Paired comparisons at nominal 90% coverage. The comparator is invariant linear Ad-EffOrt for AR(1), and the invariant nonlinear RF-center method for variable coefficients. Length ratios are IN-ARCP divided by comparator. Bounds use 3.890592 Monte Carlo standard errors.

Scenario	m	IN-ARCP coverage	Comparator coverage	Length ratio	Ratio interval
AR(1)	4	0.8994	0.9000	0.863	[0.855, 0.871]
AR(1)	12	0.9003	0.9002	0.869	[0.863, 0.874]
AR(1), scale x2	4	0.9013	0.9003	0.866	[0.858, 0.873]
AR(1), scale x2	12	0.9001	0.9003	0.867	[0.860, 0.873]
Variable coefficient	4	0.9001	0.9001	1.186	[1.175, 1.196]
Variable coefficient	12	0.8996	0.8992	1.308	[1.298, 1.318]

TABLE 4. Equal-fit classical comparison. C denotes IN-ARCP; S denotes the Gaussian plug-in Student reference.

Scenario	m	C coverage	S coverage	Length C/S
Gaussian AR(1)	4	0.8994	0.9001	0.998
Gaussian AR(1)	12	0.9003	0.9001	1.002
AR(1), t5 innovations	4	0.8994	0.8886	1.055
AR(1), t5 innovations	12	0.9001	0.8947	1.024

E. BOUNDARY OF SCALE-TRANSFER VALIDITY

Fig. 6 separates whole-episode scale transfer from standardized-shape changes. Innovation normalization preserves coverage under whole-episode doubling in the common-AR setting. A coefficient change or future-only scale multiplication does not satisfy Proposition 1. In a separate fresh fitted validation experiment, future-only multiplication gives coverage approximately 0.573 for $m = 4$ and 0.526 for $m = 12$. Thus the simulated coverage loss occurs under changes for which Proposition 1 provides no guarantee.

The separate validation includes 4000 fresh fitted repetitions across the assessed cases, with 800 devoted to future-only negative controls. It is separate from the 4800-repetition comparator study. These experiments evaluate the stated synthetic distributions.

VIII. DISCUSSION

A. TRAINING-SCALE HETEROGENEITY AND INTERVAL LENGTH

The equations account for three sources of variation in interval length: the estimated coefficient, the calibration threshold and the observed test history. Calculating average length at a fixed coefficient leaves out uncertainty from training. Analyzing coefficient error alone leaves out calibration. Equations (13)–(16) account for both, while (22) separates their leading effects when the training and calibration sets are large.

The analysis uses established results from normalized conformal prediction, expected set size and invariant prediction. Its specific result is the average-length calculation and its explicit approximation for the autoregressive model. The factor κ shows why removing amplitude from the prediction score does not remove its effect on learning the coefficient: recordings with larger amplitudes receive greater weight in

the pooled estimator.

B. MODEL STRUCTURE AND COMPARATIVE EFFICIENCY

Under common Gaussian dynamics, the innovation transform produces an independent Gaussian history and future residual, exposing the Student pivot. A more general learned scale must estimate structure that is available analytically in this model. This explains why the simple construction can outperform a specified invariant learned comparator while remaining close to the classical reference. It does not show that the broader learner is intrinsically deficient.

The unmodified comparators can use absolute amplitudes when fitting to the training distribution. Our adaptations instead normalize each recording by its history amplitude, so multiplying a recording by a positive constant multiplies its interval endpoints by the same constant. These procedures answer different questions and are reported separately. The 13–14% reduction is relative to the adapted Ad-EffOrt comparator. The separate Gaussian Student reference uses the fitted coefficient; the analytical oracle uses the true coefficient. The much smaller cost of coefficient estimation does not explain that entire difference.

With variable coefficients, a shared linear predictor has limited ability to represent episode-specific dynamics. A nonlinear center can use the history to adapt its prediction; the observed advantage of the RF-center procedure is consistent with this explanation, although the procedures also differ in their scale construction. The rank guarantee can remain applicable when the standardized mixture is common across splits, even though the common-Gaussian-coefficient efficiency equations no longer apply.

IX. PRACTICAL INTERPRETATION, LIMITATIONS AND FUTURE WORK

A. PRESENT USE AS AN ANALYTICAL BENCHMARK

The immediate use of the analysis is to study the efficiency cost of estimating and calibrating a predictor for short dependent histories. The known-coefficient pivot supplies a reference, the finite-calibration expression isolates order-statistic cost, and the fitted-mean characterization incorporates training randomness. Together they provide a benchmark against which more flexible procedures can be evaluated under a controlled observation model.

A possible use is next-reading uncertainty from short, independent sensor recordings whose standardized dynamics

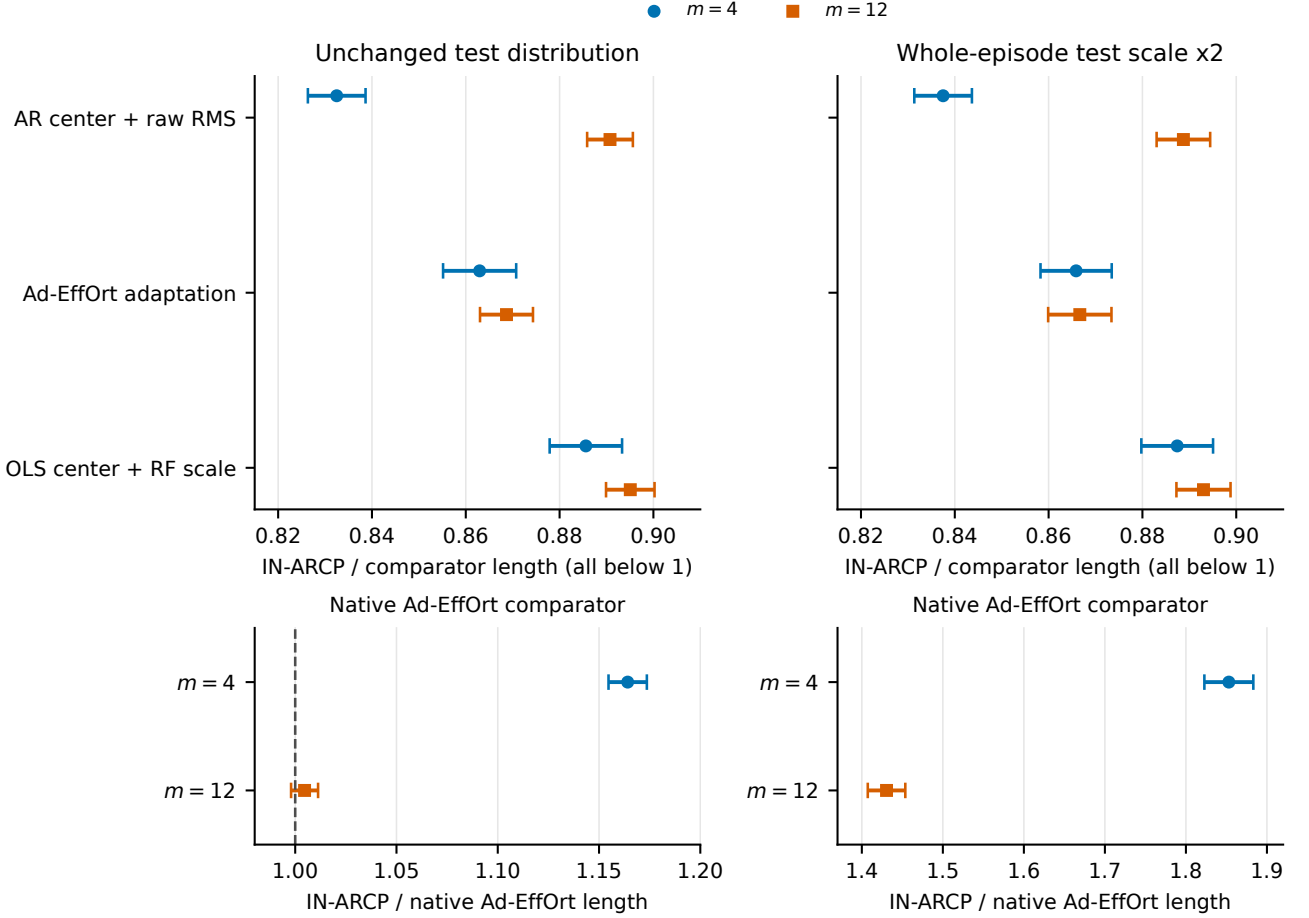


FIGURE 4. Mean-length ratios for common Gaussian AR(1) episodes, with unchanged (left) and doubled (right) test amplitude. The numerator is IN-ARCP length. Top panels enlarge the 0.815–0.910 range for the raw-RMS ablation and scale-invariant adaptations; every displayed ratio is below one. Bottom panels show the native Ad-EffOort comparison on separate, explicitly labeled scales. Circles denote $m = 4$, squares $m = 12$. Bars use (23) with multiplier 3.890592. The dashed line marks equal length where it lies within the displayed range. Native Ad-EffOort undercovers after scale shift; its shorter intervals there are not a matched-coverage efficiency gain.

are stable while their amplitudes vary. Before applying the construction, one must identify the scalar observable and its physical units, define independent recording units, obtain any centering parameters without using future responses, and separate fitting from calibration. Overlapping windows from one long record are not automatically independent episodes. The output is an interval with a specified marginal coverage statement, not guaranteed accuracy on every realized history.

B. MODEL AND EXPERIMENTAL LIMITATIONS

Several technical limits remain. Exact distributional equations require the stationary Gaussian common-coefficient model. The broader coverage proposition requires exchangeable standardized calibration and test episodes; a shared mixture of dynamics can satisfy that premise, whereas arbitrary changes between splits need not. Nonzero or estimated means, unknown model order, cross-episode dependence and observation errors are not analyzed. The oracle restriction to scale-equivariant intervals is essential. The

asymptotic expansion requires the stated scale moments and an interior coefficient, and has no explicit finite-sample error bound. The complete fitted-mean numerical check covers one configuration; integration over random training scales and broad finite-sample accuracy remain untested.

The empirical ranking is restricted to the tested generators, nominal level, sample sizes and implementations. The scale-normalized versions of Ad-EffOort and locally weighted regression differ from their unmodified forms. No numerical ranking against CD-split, HPD-split, SymmPI or conformalized Kalman filters is available. Tuning budgets were fixed, and comprehensive matched-budget sensitivity has not been established. No runtime benchmark or real-data validation is reported.

C. DATA ALLOCATION AND SCALE-BALANCED ESTIMATION

The most direct extension is experimental design for independent episodes. With a fixed acquisition budget, the expansion suggests investigating the allocation of episodes

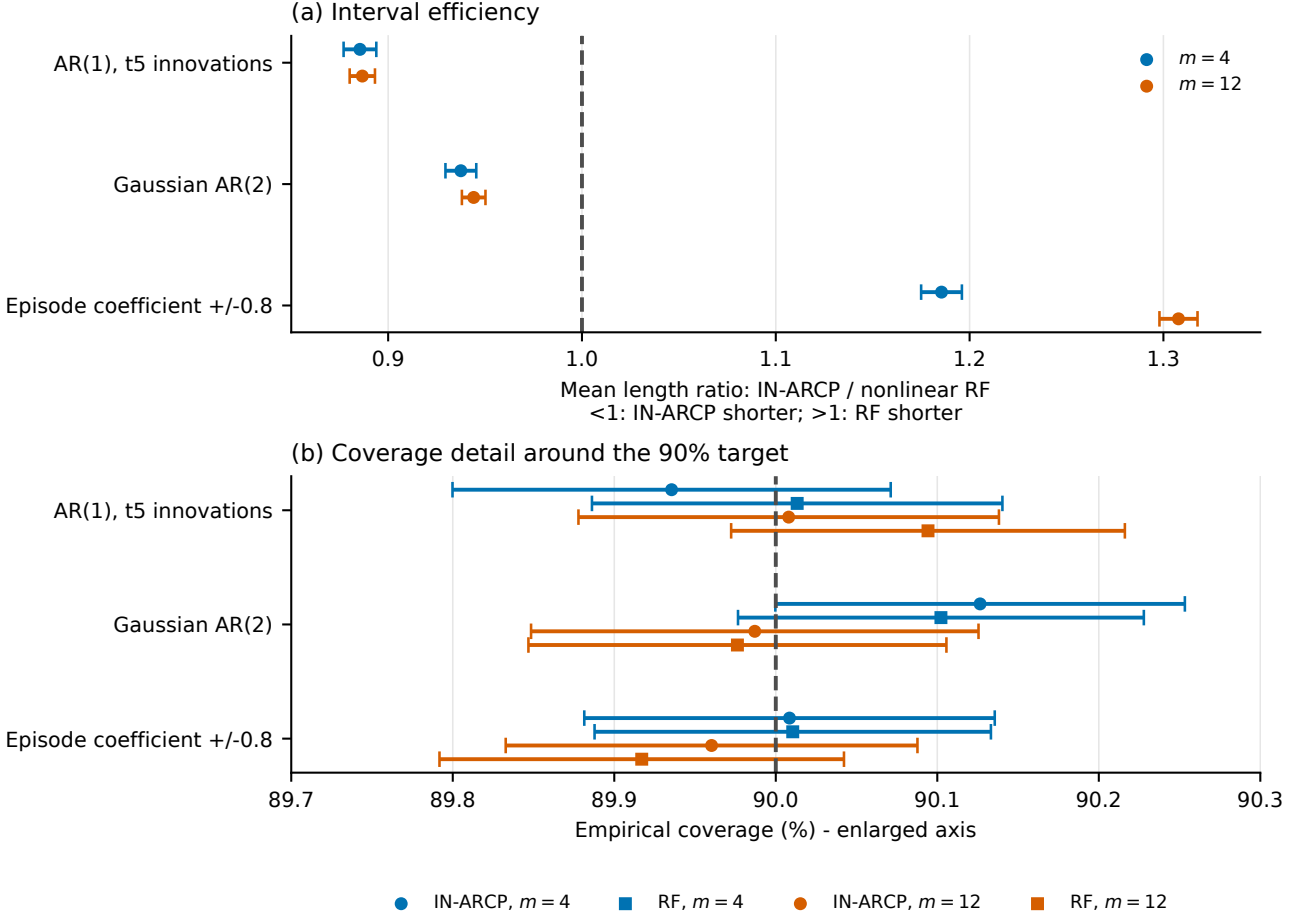


FIGURE 5. Structural challenges with doubled test amplitude. (a) Paired length ratios versus the invariant nonlinear RF-center method, using multiplier 3.890592. (b) Coverage in percent with approximate 95% Monte Carlo intervals, on an explicitly enlarged 89.7–90.3% axis. The dashed lines mark equal length and nominal 90% coverage, respectively. Blue denotes $m = 4$, orange $m = 12$; in (b), circles denote IN-ARCP and squares the RF-center procedure. Variable dynamics favor the nonlinear comparator.

between fitting and calibration. A useful allocation rule would need to account for quantile rounding and be checked against the finite-sample mean across N , n , m and κ ; leading terms alone do not establish an optimal finite-sample split.

The training heterogeneity factor also motivates scale-balanced estimation. Such an estimator would require its own bias and variance analysis: normalizing a training episode changes the estimating equation and can introduce dependence between its weights and residuals. A meaningful comparison would hold the calibration procedure fixed and assess mean length at matched coverage. The variable-dynamics results separately motivate flexible centers or regime-specific models, with selection handled by an appropriate validity-preserving procedure [16].

D. FUTURE APPLICATION TO CIVILIAN RADAR MEASUREMENTS

A possible future test uses the next-scene median motion-compensated radial velocity of detections labeled as static

surroundings in RadarScenes [19]. Annotation-based selection makes this a retrospective scalar-prediction task. Independent recordings, causal centering and an observation model for changing scene composition would be needed before testing the present assumptions. Evaluation should compare persistence, affine AR and flexible predictors using coverage and width in velocity units. A separate measurement-quality criterion would be needed to establish practical value. This application requires validation on measurements. Prediction coverage alone would not establish detection probability, false-alarm control or interference rejection. Latent-state tracking would require a state-space model and trajectory-aware calibration, as in [15]; the present interval predicts an observed value.

X. CONCLUSION

For independent autoregressive recordings, the analysis separates maintaining prediction coverage from controlling average interval length. The integral expression accounts for coefficient estimation, calibration and the random test

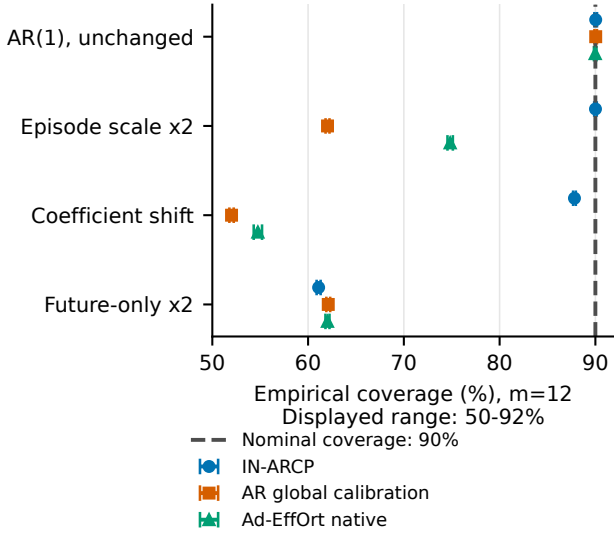


FIGURE 6. Coverage in the main comparison at history length $m = 12$, displayed over 50–92% to resolve the observed failures. Bars are approximate 95% Monte Carlo intervals; some are smaller than the markers. The dashed line is the nominal 90% target. Whole-episode doubling preserves the normalized-score premise; coefficient and future-only changes do not. The displayed range is enlarged, not the full probability scale.

history. Its large-sample approximation shows how amplitude variation in training data and rounding the calibration rank affect length. Numerical checks support the individual equations and one complete average-length calculation. The comparisons show shorter intervals than specified adapted methods under common dynamics, but longer intervals than a nonlinear comparator when dynamics vary. IN-ARCP provides a tractable reference for studying training and calibration costs in amplitude-varying autoregressive episodes.

APPENDIX A DERIVATION OF THE EFFICIENCY EXPANSION

A. CURVATURE AT THE CORRECTLY SPECIFIED COEFFICIENT

Let $q(r) = G_r^{-1}(p)$ and $h(r) = q(r)J(r)/c$. Write $r = \rho + \delta$, $A = L^T Q'_\rho L$, and

$$\begin{aligned} b_r(D) &= 1 + \delta u(D) + \delta^2 v(D) + O(\delta^3), \\ a_r(D) &= \delta a_1(D), \quad q(r) = c + \delta q_1 + \delta^2 q_2 + O(\delta^3), \end{aligned}$$

where $u(D) = D^T A D / 2$. Smoothness holds locally because Q_r is positive definite and $G'_\rho(c) = 2f_m(c) > 0$. Expanding (10) at fixed coverage gives

$$\begin{aligned} q_1 &= -c \mathbb{E}u, \\ q_2 + q_1 \mathbb{E}u + c \mathbb{E}v &= -\frac{f'_m(c)}{2f_m(c)} \{c^2 \text{Var}(u) + \mathbb{E}a_1^2\}. \end{aligned} \quad (24)$$

The left side is the quadratic coefficient of $q(r)J(r)$; its linear coefficient vanishes. Uniform-sphere moments give

$$\begin{aligned} \text{Var}(u) &= \frac{\text{tr}(A^2) - \text{tr}(A)^2/m}{2m(m+2)}, \\ \mathbb{E}a_1^2 &= \|e_m^T L\|^2 = d^{-1}. \end{aligned} \quad (25)$$

The tridiagonal Q_r has endpoint diagonal entries one, interior entries $1 + r^2$, and off-diagonal entries $-r$. Its determinant is $1 - r^2$. Differentiating its log determinant yields

$$\begin{aligned} \text{tr}(A) &= -2\rho/d, \\ \text{tr}(A^2) &= 2(m-2)/d + 2(1 + \rho^2)/d^2. \end{aligned} \quad (26)$$

Substitution gives (19). Finally, $f'_m(c)/f_m(c) = -(m+1)c/(m+c^2)$, so

$$h(\rho + \delta) = 1 + K\delta^2 + O(\delta^3), \quad (27)$$

with K in (20).

B. TRAINING CONTRIBUTION AND EXPECTATION CONTROL

For a training episode let $S_i = \sum_{j=1}^m X_{ij} \varepsilon_{i,j+1}$ and $D_i = \sum_{j=1}^m X_{ij}^2$. Then

$$\tilde{\rho} - \rho = \frac{\sum_i S_i}{\sum_i D_i}, \quad \mathbb{E}S_i = 0, \quad \mathbb{E}D_i = m\mathbb{E}\sigma^2. \quad (28)$$

Cross-transition terms in the numerator have zero covariance by conditioning on the past innovation history. Hence $\text{Var}(S_i) = m d \mathbb{E}\sigma^4$, and the iid episode central limit theorem and ratio expansion give

$$\sqrt{N}(\hat{\rho} - \rho) \implies \mathcal{N}(0, V_\rho). \quad (29)$$

Clipping does not affect this limit because the true coefficient lies in the interior. To pass from distributional convergence to the mean expansion, choose a sufficiently small $\delta_0 > 0$ such that $\mathbb{E}|S_i|^{2+\delta_0} < \infty$, implied by the extra scale moment. Since $D_i > 0$ almost surely and $\mathbb{E}D_i > 0$, a lower-tail Chernoff bound controls $\sum_i D_i < N\mathbb{E}D_i/2$ exponentially. On its complement, moment bounds for sums of centered S_i give uniform integrability of $N(\hat{\rho} - \rho)^2$. Clipping bounds the exceptional-event contribution. Thus

$$N\mathbb{E}(\hat{\rho} - \rho)^2 \rightarrow V_\rho, \quad \mathbb{E}h(\hat{\rho}) = 1 + KV_\rho/N + o(N^{-1}). \quad (30)$$

The same moment control makes the contribution away from a fixed neighborhood of ρ smaller than order N^{-1} , justifying use of the local expansion inside the expectation.

C. CALIBRATION AND QUANTILE ROUNDING

At the true coefficient define $g(u) = F_m^{-1}((1+u)/2)$. Then

$$g'(p) = \frac{1}{2f}, \quad g''(p) = -\frac{f'_m(c)}{4f^3}. \quad (31)$$

The beta variable in (11) satisfies

$$\begin{aligned} \mathbb{E}B &= p + \epsilon_n/(n+1), \\ \text{Var}(B) &= p(1-p)/(n+2) + O(n^{-2}). \end{aligned} \quad (32)$$

A second-order expansion of $\mathbb{E}g(B)/c$ gives the calibration and rounding terms in (22). For fitted r , the corresponding local coefficients are continuous on the compact clipped range and converge to their values at ρ . Their expectation errors are $o(n^{-1})$.

To control the beta-integral remainder, eigenvalues of the history quadratic form remain bounded away from zero on the clipped range, and the normalized center displacement is bounded. Consequently inverse-score quantiles grow at most as a constant times $1 + (1 - u)^{-1/m}$, uniformly in that range. Derivatives are bounded in a fixed neighborhood of p ; beta concentration, including its endpoint power, controls the complement. Combining this with the training expansion proves Theorem 1, without requiring a fixed ratio N/n .

APPENDIX B NUMERICAL EVALUATION OF DISTRIBUTIONAL INTEGRALS

An independent representation uses a Gaussian quadratic form. Let C be the Cholesky factor of the correlation matrix with entries $\rho^{|j-l|}$, for an entire episode. Put

$$\begin{aligned} Q_r &= W_r^T W_r, \quad v_r = (0, \dots, 0, -r, 1)^T, \\ B_r(q) &= C^T \{v_r v_r^T - q^2 \text{diag}(Q_r/m, 0)\} C. \end{aligned} \quad (33)$$

If $\lambda_l(q)$ are its eigenvalues, then

$$\begin{aligned} \phi_{r,q}(t) &= \prod_{l=1}^{m+1} (1 - 2it\lambda_l(q))^{-1/2}, \\ G_r(q) &= \frac{1}{2} - \frac{1}{\pi} \int_0^\infty \frac{\text{Im} \phi_{r,q}(t)}{t} dt. \end{aligned} \quad (34)$$

The complex branches are continuous from $\phi_{r,q}(0) = 1$. The score event is precisely that this quadratic form is non-positive; characteristic-function inversion [14] gives (34).

The radius expectation can also be evaluated without sphere sampling. If η_j are the eigenvalues of $C_H^T Q_r C_H/m$, where C_H is the history correlation Cholesky factor, then at unit marginal scale

$$\mathbb{E}s_r(H) = \frac{1}{\sqrt{\pi}} \int_0^\infty \frac{1 - \prod_{j=1}^m (1 + 2u^2\eta_j)^{-1/2}}{u^2} du. \quad (35)$$

This follows from the Laplace-transform representation of the square root. Equations (34) and (35) provide a separate route for checking (13).

CODE AND DATA AVAILABILITY

The IN-ARCP Python implementation, usage examples and tests are available under the MIT license [20]. Simulation configurations, numerical tolerances and reference outputs are provided in the accompanying reproducibility materials. The study uses synthetic data.

ACKNOWLEDGMENT AND DISCLOSURE

OpenAI Codex [21] assisted with the mathematical exposition, simulation and verification code, literature organization, and drafting of the abstract, Sections I–X and the appendices. The use of this system included generation and revision of research prose and code.

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